

Towards Secure Container Infrastructure on RISC-V: The Development from RustVMM to Kata-Containers

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Kata-Containers Maintainer

Cloud-Hypervisor Maintainer

Rust-VMM Infra Maintainer

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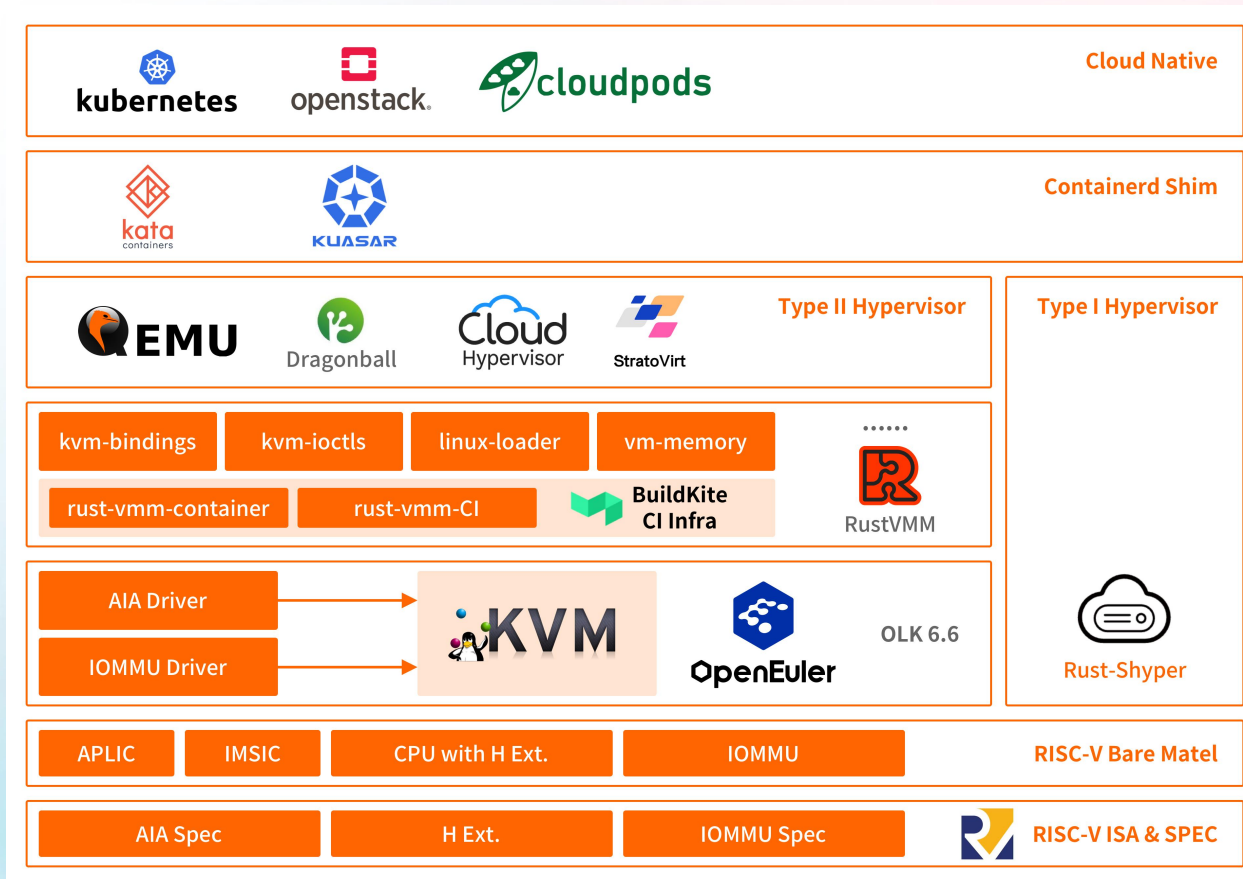
01 Roadmap



Roadmap — Stage One

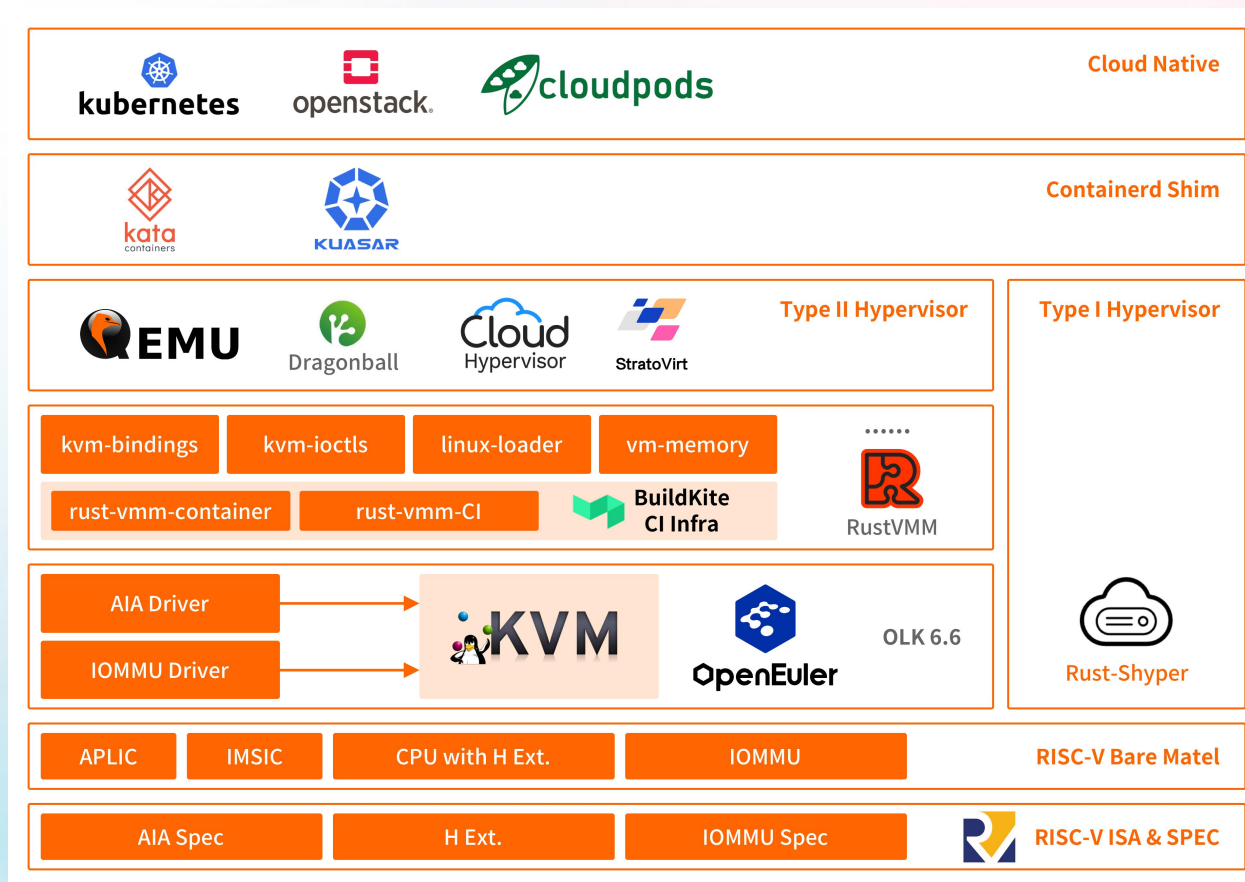
A bottom up, usable, maintainable and dependable software stack on RISC-V platforms.

- Orchestrated by Kubernetes
- Secured by Kata-Containers
- Speedup by Cloud-Hypervisor
- Powered by RustVMM
- On openEuler RISC-V



Roadmap – Stage Two

A fully supported, reliable and prosperous virtualization software ecosystem of RISC-V.
Enabling RISC-V to compete with x86 and ARM in both Virtualization & Cloud Native software ecosystem.



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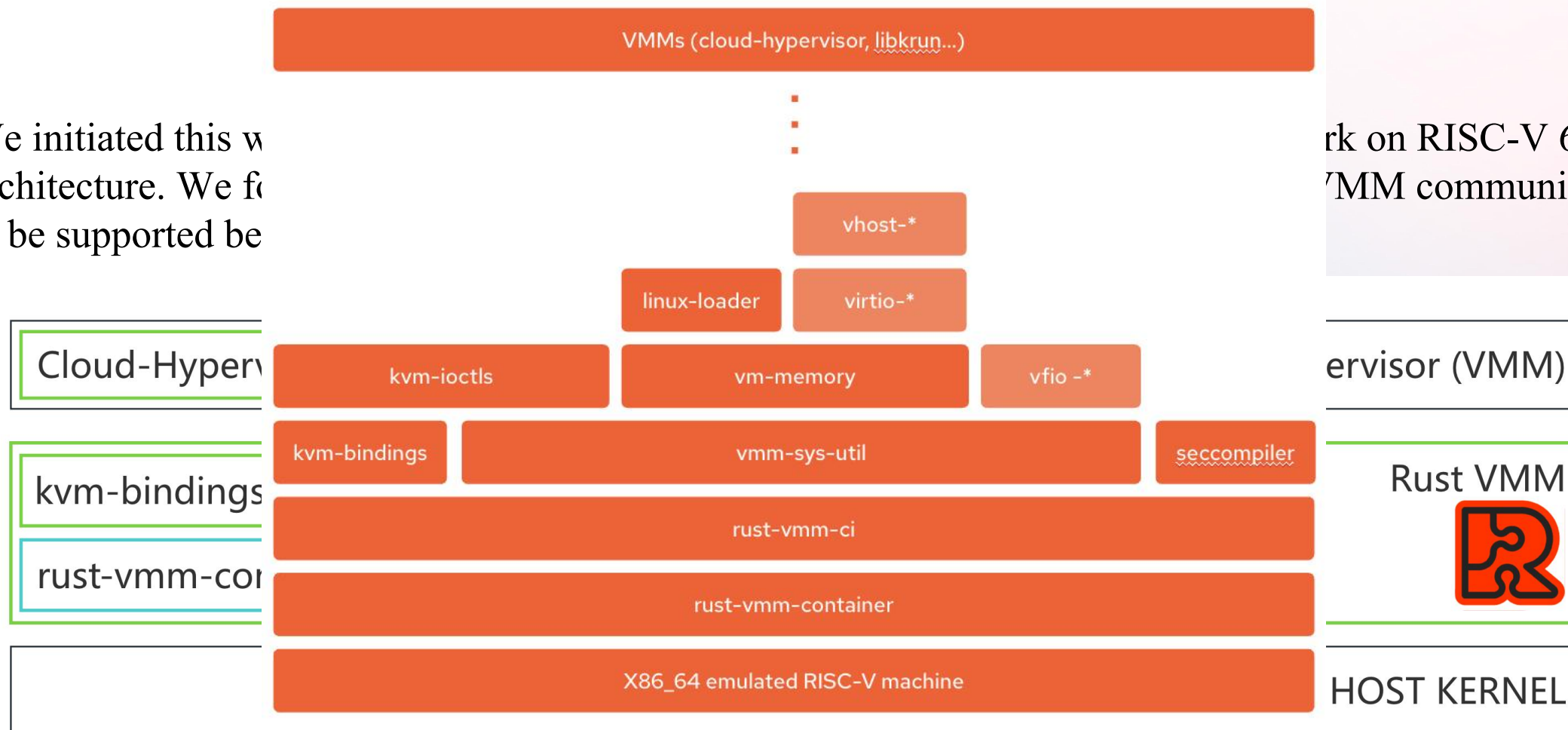
02 Work Progress



Work Proaress – RustVMM

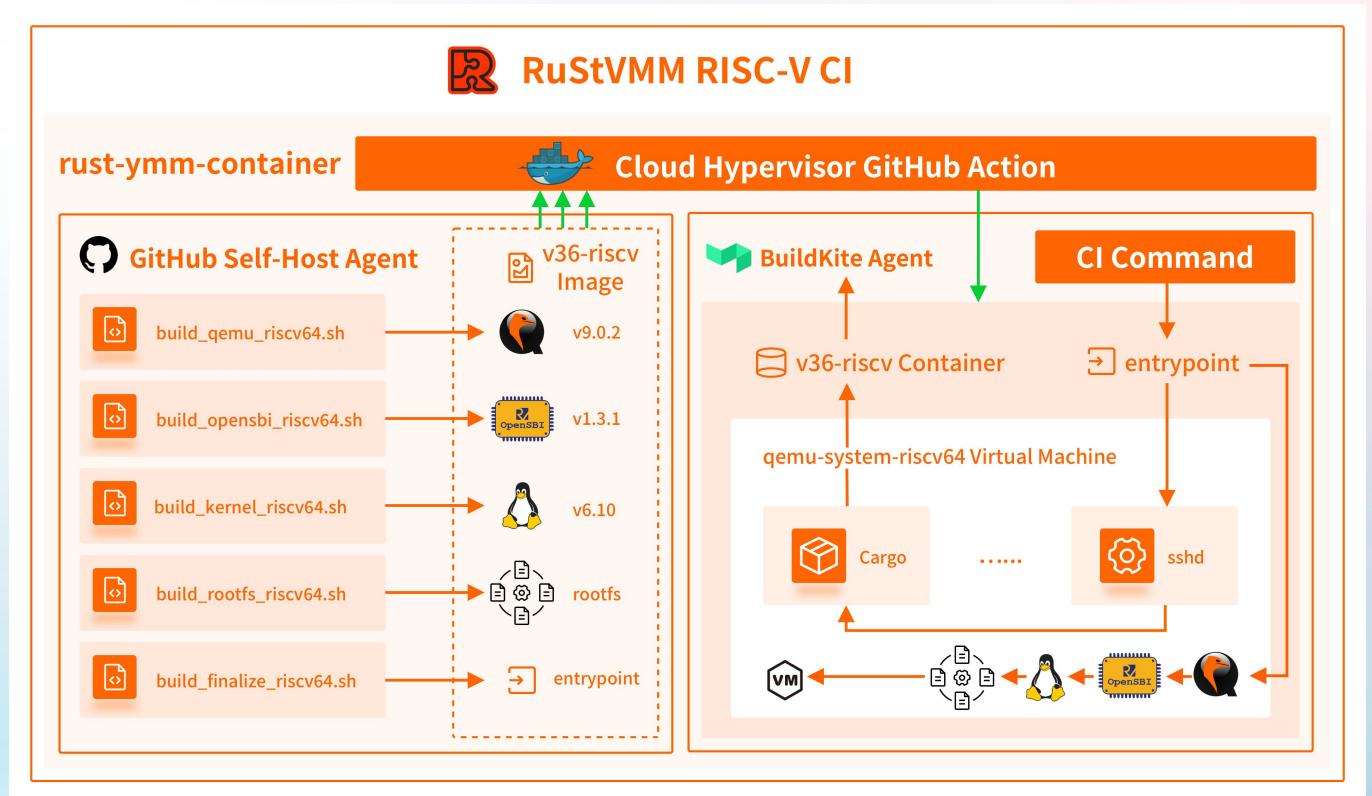
We initiated this w
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rk on RISC-V 64-bit
VMM community need



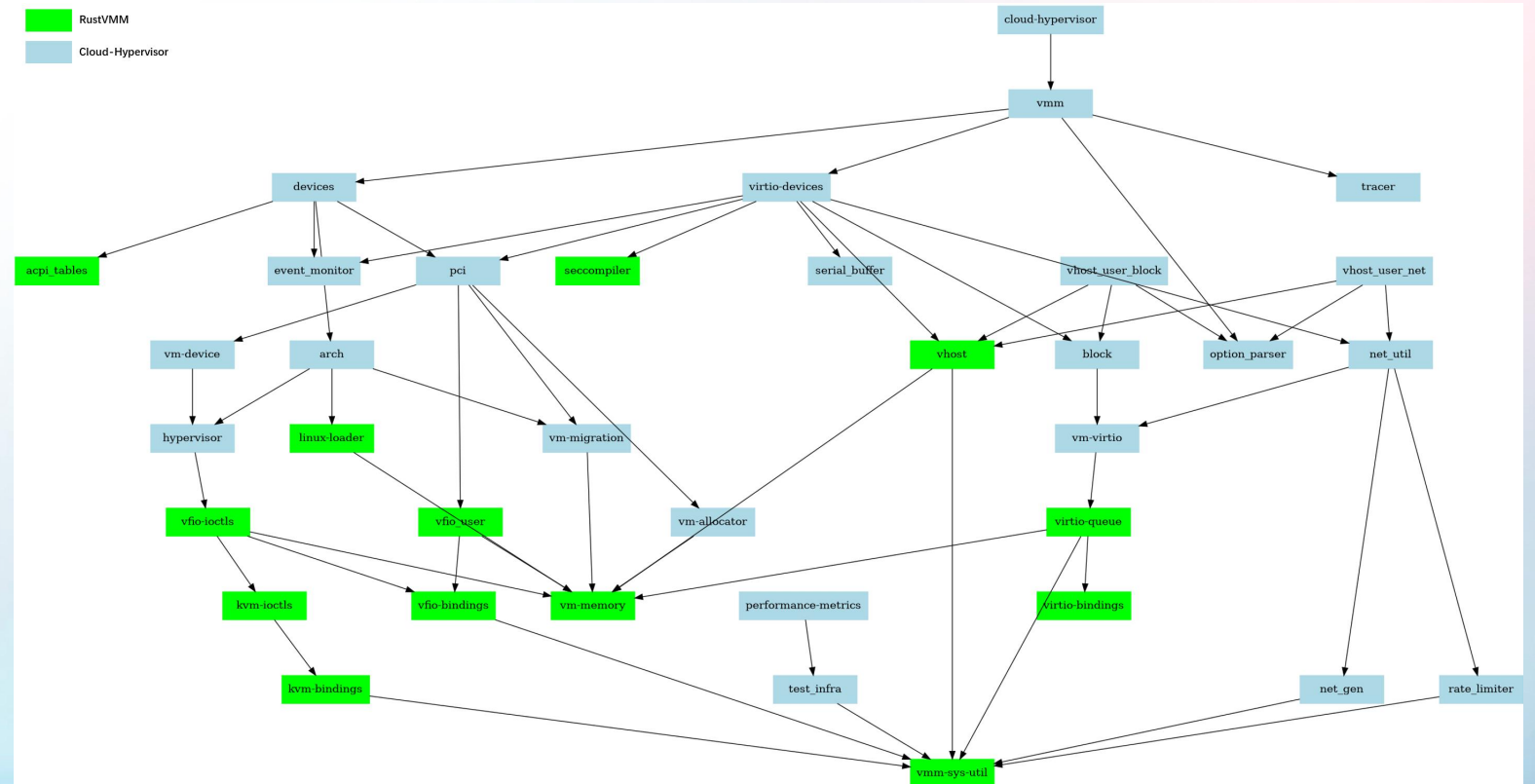
Work Progress – RustVMM

As aforementioned that the actual hardware is not yet made available for us, how the CI should be designed to look after RISC-V code becomes a problem.

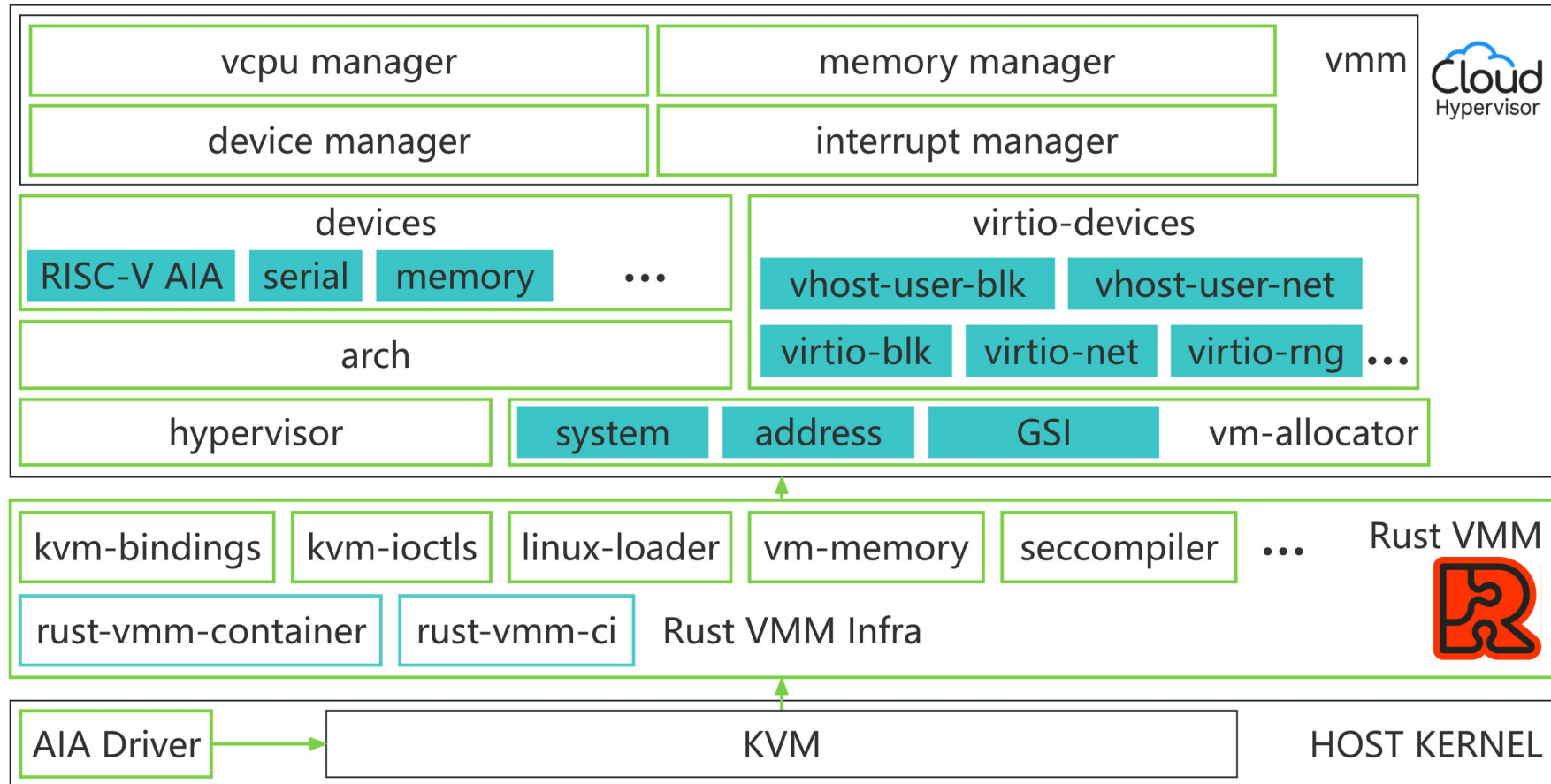


Work Progress — Cloud-Hypervisor

This is a dependency graph of Cloud-Hypervisor, which shows the inter-connection of crates of Cloud-Hypervisor and RustVMM.

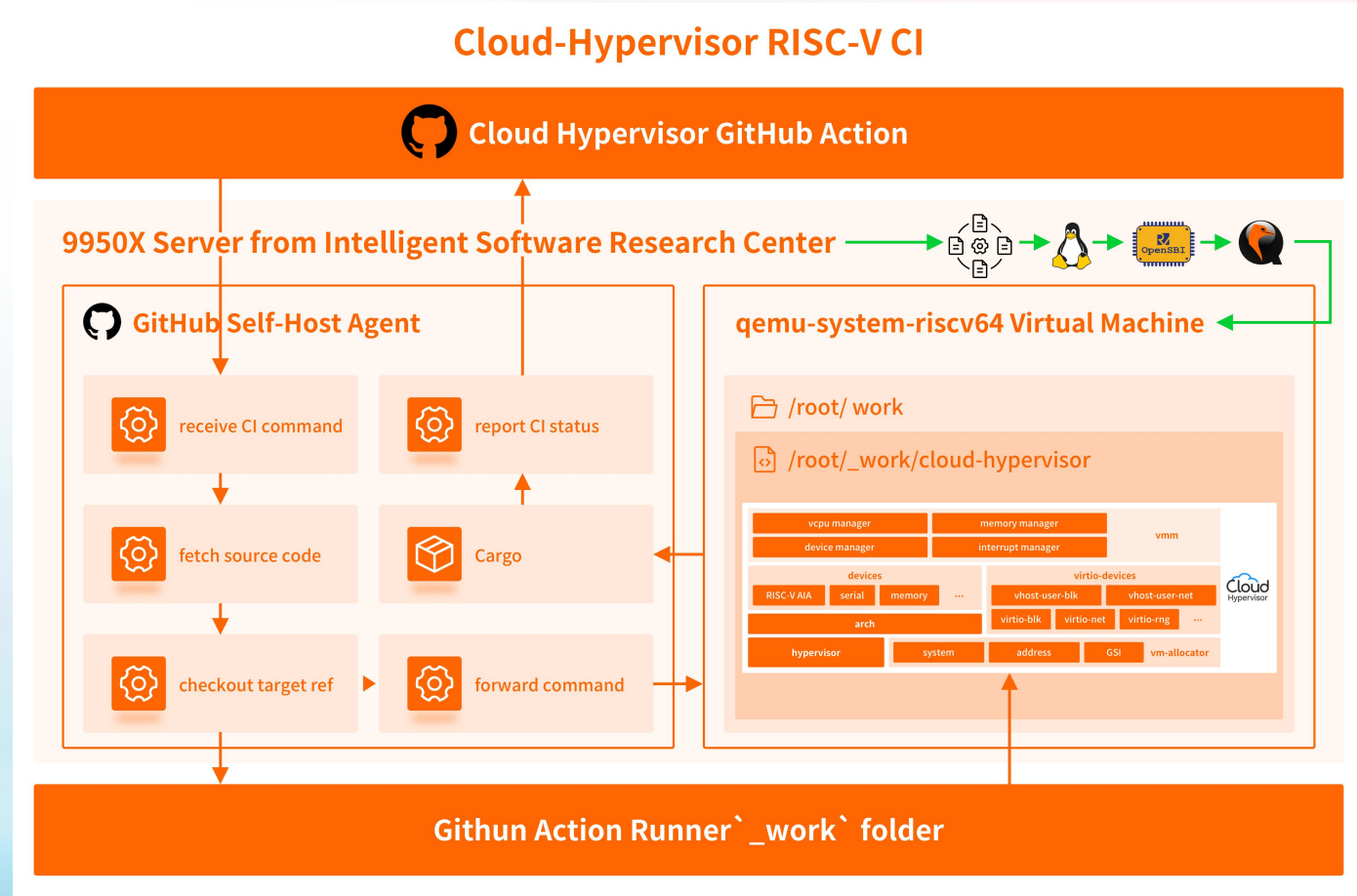


Stage 1 RISC-V Support – Cloud-Hypervisor



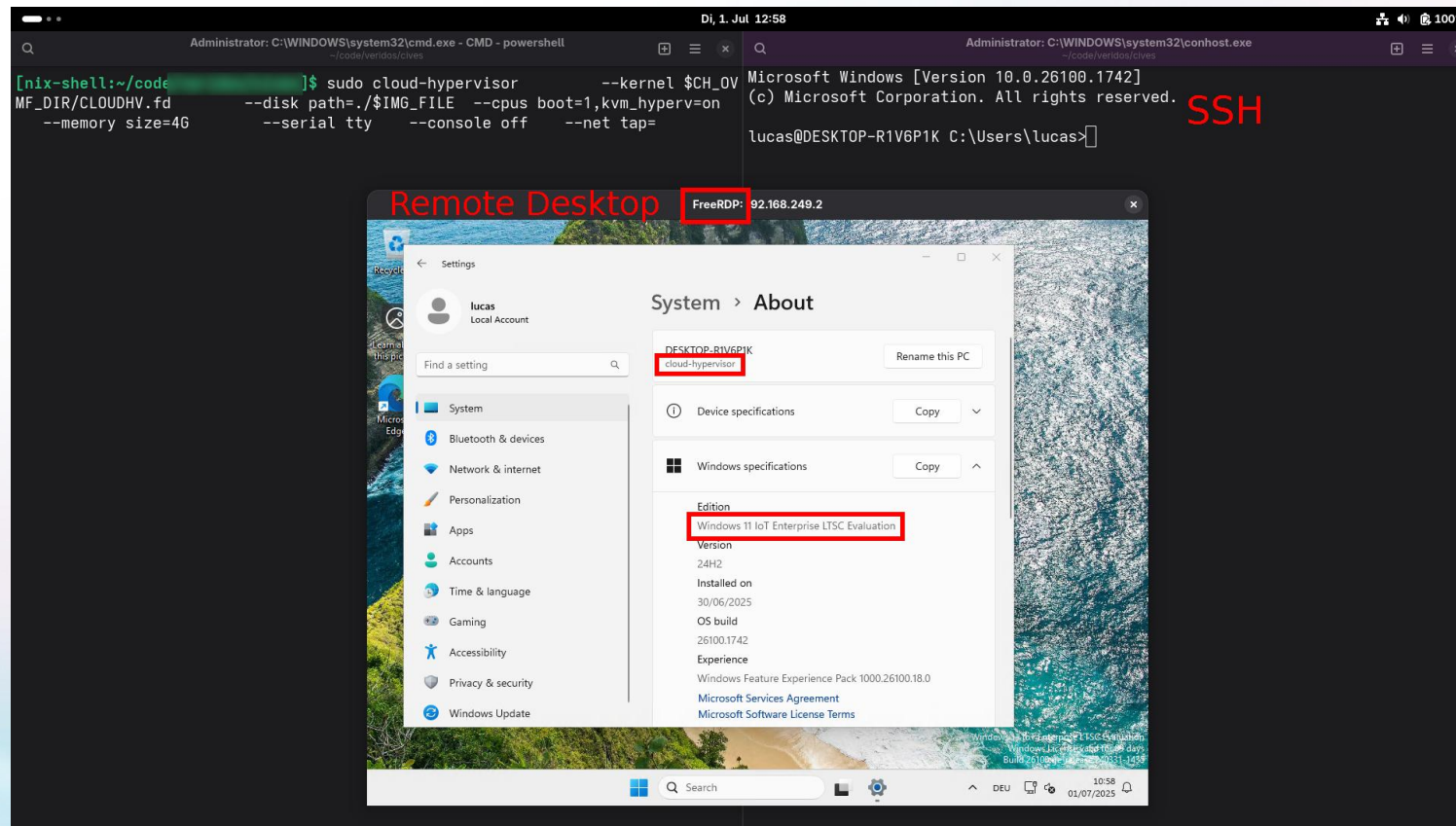
CI Status — Cloud-Hypervisor

Here again, CI for H extension based software like Cloud-Hypervisor need special design and handling. We adopted similar approach used in RustVMM community.



Stage 2 RISC-V Support — Cloud-Hypervisor

Cloud-Hypervisor v45 and onwards now supports direct boot of RISC-V Linux, but there are still many feature gaps.



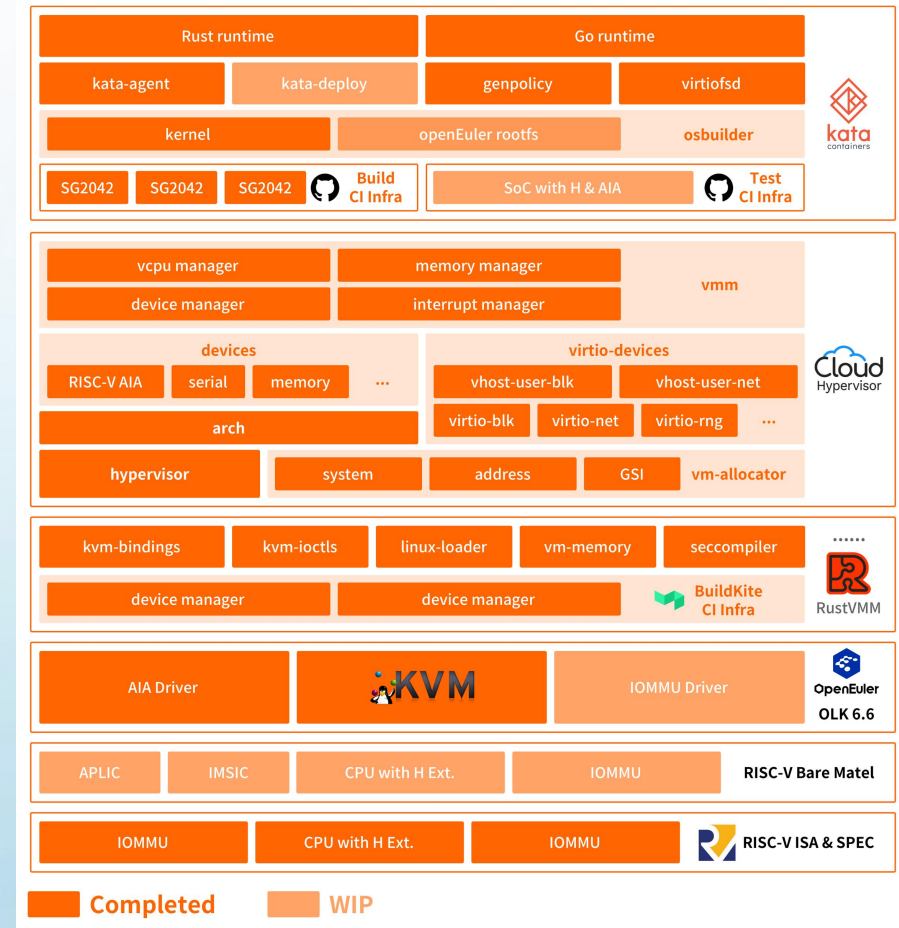
Work Progress — Kata-Containers

We have now worked out a full Rust-based secure container software stack, RISC-V support for:

- Go runtime
- Rust runtime
- Agent

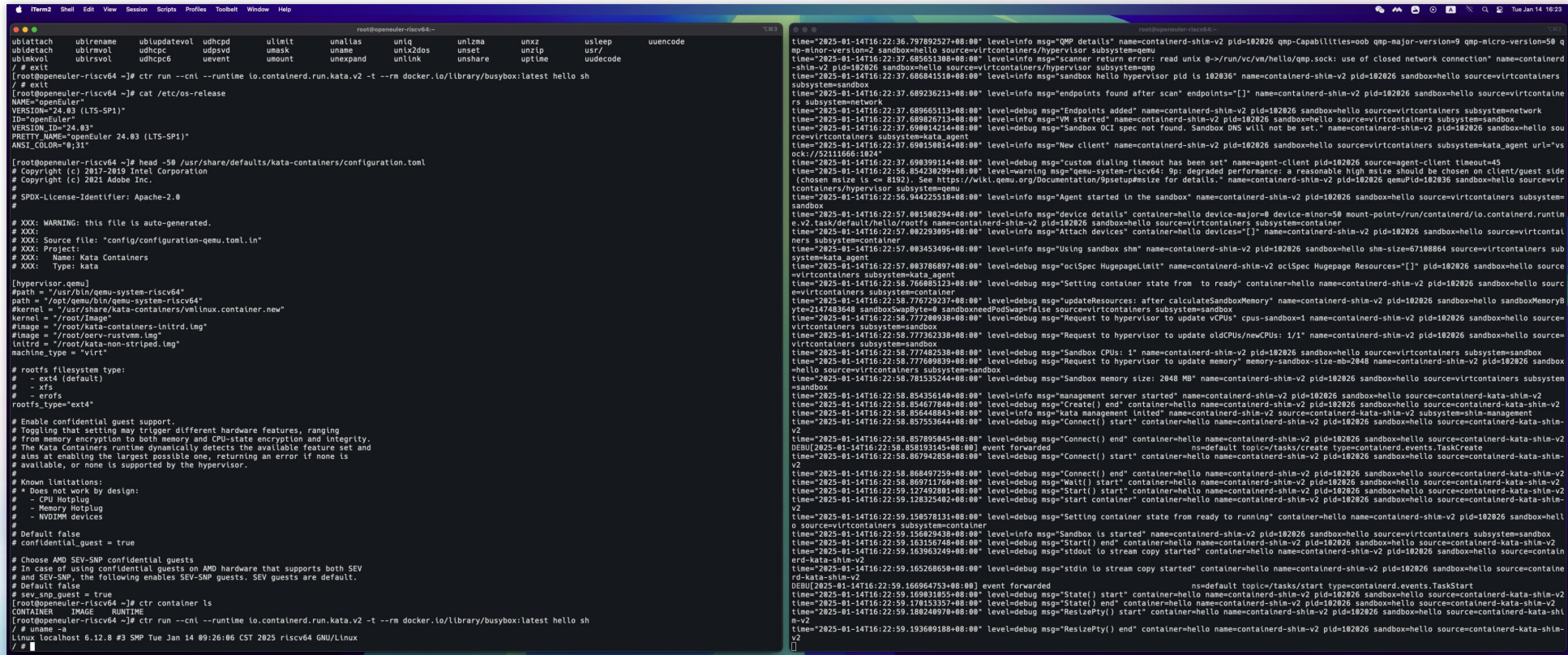
are completed, released in 3.17.0.

Based on RISC-V CI Infra for Kata, with real RISC-V hardware (SG2042) for building.



Work Progress — Kata-Containers

We have now successfully launched v3.13.0 Kata-Containers on openEuler RISC-V 24.03 LTS SP1 with its go runtime and QEMU v8.2.0 as its hypervisor.



```
root@openEuler-riscv64:~# cat /etc/os-release
NAME="openEuler"
VERSION="24.03 (LTS-SP1)"
ID="openEuler"
VERSION_ID="24.03"
PRETTY_NAME="openEuler 24.03 (LTS-SP1)"
ANSI_COLOR="0;31"

root@openEuler-riscv64:~# head -50 /usr/share/defaults/kata-containers/configuration.toml
# Copyright (c) 2017-2019 Intel Corporation
# Copyright (c) 2021 Adobe Inc.
# SPDX-License-Identifier: Apache-2.0

# XXX: WARNING: this file is auto-generated.
# XXX:
# XXX: Source file: "config/configuration-qemu.toml.in"
# XXX: Project:
# XXX: Name: Kata Containers
# XXX: Type: kata

[hypervisor.qemu]
#path = "/usr/bin/qemu-system-riscv64"
#path = "/opt/qemu/bin/qemu-system-riscv64"
#kernel = "/usr/share/kata-containers/vmlinux.container.new"
#kernel = "/root/Image"
#image = "/root/kata-containers-initrd.img"
#image = "/root/ovm-rustvm.img"
#initrd = "/root/kata-non-striped.img"
#machine_type = "virt"

# rootfs filesystem type:
# - ext4 (default)
# - xfs
# - erofs
#rootfs.type="ext4"

# Enable confidential guest support.
# Toggling that setting may trigger different hardware features, ranging
# from memory encryption to both memory and CPU-state encryption and integrity.
# The Kata Containers runtime dynamically detects the available feature set and
# aims at enabling the largest possible one, returning an error if none is
# available, or none is supported by the hypervisor.

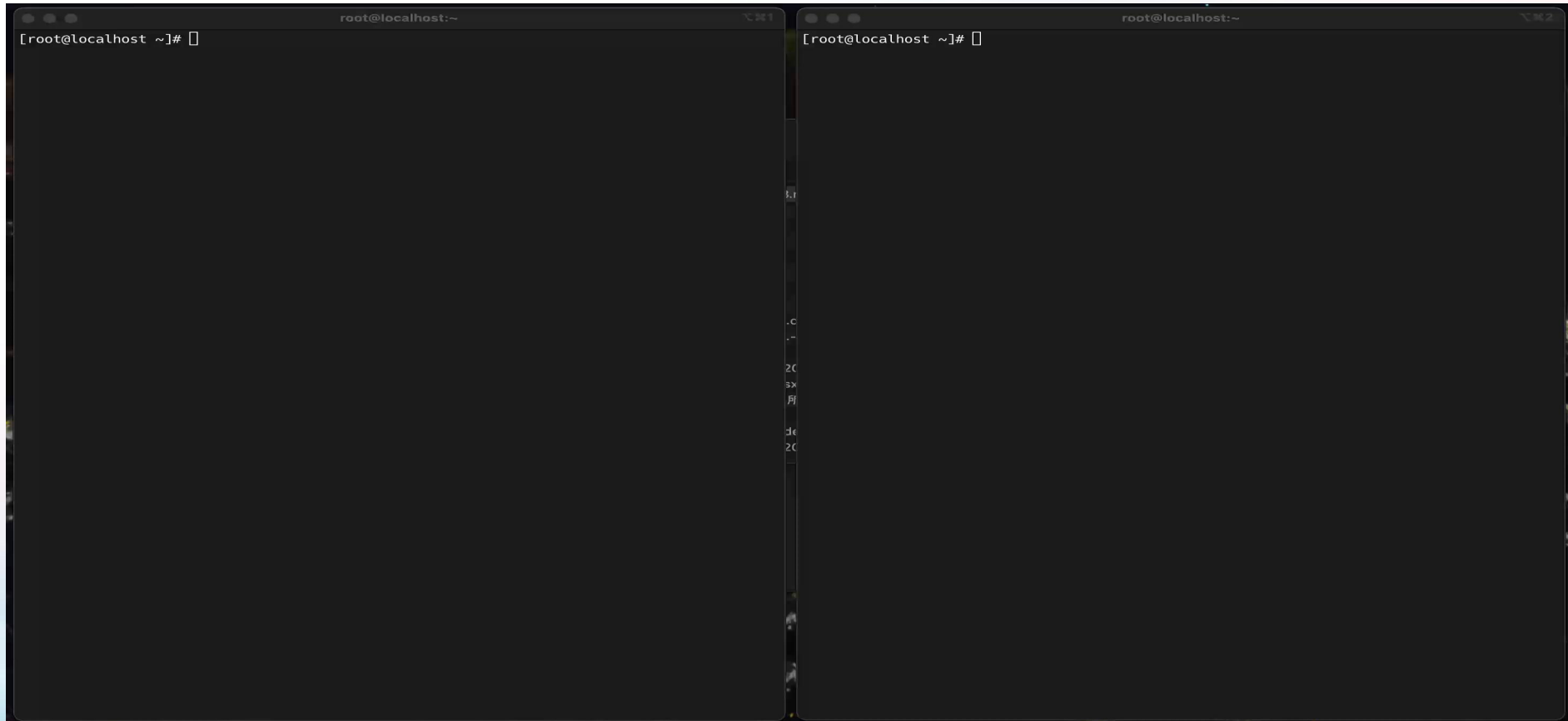
# Known limitations:
# - Does not work by design:
#   - CPU Hotplug
#   - Memory Hotplug
#   - NVMM devices

# Default false
#confidential_guest = true

# Choose AMD SEV-SNP confidential guests
# In case of using confidential guests on AMD hardware that supports both SEV
# and SEV-SNP, the following enables SEV-SNP guests. SEV guests are default.
# Default false
#sev.snp_guest = true
root@openEuler-riscv64:~# ctr run --runtim io.containerd.run.kata.v2 -t --rm docker.io/library/busybox:latest hello sh
# #
Linux localhost 6.12.8 #3 SMP Tue Jan 14 09:26:06 CST 2025 riscv64 GNU/Linux
```


Demo — Kata-Containers

Invoking Kata with ctr on openEuler 24.03 LTS SP2 to start a ubuntu container image:





Thanks for Listening